Neural Computation -- 1991

55 article(s), all issues

1. Removing time variation with the anti-hebbian differential synapse

MitchisonGraeme

(no DOI)

I describe a local synaptic learning rule that can be used to remove the effects of certain types of systematic temporal variation in the inputs to a unit. According to this rule, changes in synapt...

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2. Deciphering the Brain's Codes

Masakazu Konishi

https://doi.org/10.1162/neco.1991.3.1.1 | pp. 1-18

The two sensory systems discussed in this review use similar algorithms for the synthesis of the neuronal selectivity for the stimulus that releases a particular behavior, although the neural circuits, the brain sites involved, and even the species are different. This stimulus selectivity emerges gradually in a neural network organized according to parallel and hierarchical design principles. The parallel channels contain lower order stations with special circuits for the creation of neuronal selectivities for different features of the stimulus. Convergence of the parallel pathways brings these selectivities together at a higher order station for the eventual synthesis of the selectivity for the whole stimulus pattern. The neurons that are selective for the stimulus are at the top of the hierarchy, and they form the interface between the sensory and motor systems or between sensory systems of different modalities. The similarities of these two systems at the level of algorithms suggest the existence of rules of signal processing that transcend different sensory systems and species of animals.

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3. Synchronization of Bursting Action Potential Discharge in a Model Network of Neocortical Neurons

Paul C. Bush, Rodney J. Douglas

https://doi.org/10.1162/neco.1991.3.1.19 | pp. 19-30

We have used the morphology derived from single horseradish peroxidase-labeled neurons, known membrane conductance properties and microanatomy to construct a model neocortical network that exhibits synchronized bursting. The network was composed of interconnected pyramidal (excitatory) neurons with different intrinsic burst frequencies, and smooth (inhibitory) neurons that provided global feedback inhibition to all of the pyramids. When the network was activated by geniculocortical afferents the burst discharges of the pyramids quickly became synchronized with zero average phase-shift. The synchronization was strongly dependent on global feedback inhibition, which acted to group the coactivated bursts generated by intracortical reexcitation. Our results suggest that the synchronized bursting observed between cortical neurons responding to coherent visual stimuli is a simple consequence of the principles of intracortical connectivity.

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4. Parallel Activation of Memories in an Oscillatory Neural Network

D. Horn, Marius Usher

https://doi.org/10.1162/neco.1991.3.1.31 | pp. 31-43

We describe a feedback neural network whose elements possess dynamic thresholds. This network has an oscillatory mode that we investigate by measuring the activities of memory patterns as functions of time. We observe spontaneous and induced transitions between the different oscillating memories. Moreover, the network exhibits pattern segmentation, by oscillating between different memories that are included as a mixture in a constant input. The efficiency of pattern segmentation decreases strongly as the number of the input memories is increased. Using oscillatory inputs we observe resonance behavior.

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5. Organization of Binocular Pathways: Modeling and Data Related to Rivalry

Sidney R. Lehky, Randolph Blake

https://doi.org/10.1162/neco.1991.3.1.44 | pp. 44-53

It is proposed that inputs to binocular cells are gated by reciprocal inhibition between neurons located either in the lateral geniculate nucleus or in layer 4 of striate cortex. The strength of inhibitory coupling in the gating circuitry is modulated by layer 6 neurons, which are the outputs of binocular matching circuitry. If binocular inputs are matched, the inhibition is modulated to be weak, leading to fused vision, whereas if the binocular inputs are unmatched, inhibition is modulated to be strong, leading to rivalrous oscillations. These proposals are buttressed by psychophysical experiments measuring the strength of adaptational aftereffects following exposure to an adapting stimulus visible only intermittently during binocular rivalry.

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6. Dynamics and Formation of Self-Organizing Maps

Jun Zhang

https://doi.org/10.1162/neco.1991.3.1.54 | pp. 54-66

Amari (1983, 1989) proposed a mathematical formulation on the self-organization of synaptic efficacies and neural response fields under the influence of external stimuli. The dynamics as well as the equilibrium properties of the cortical map were obtained analytically for neurons with binary input-output transfer functions. Here we extend this approach to neurons with arbitrary sigmoidal transfer function. Under the assumption that both the intracortical connection and the stimulus-driven thalamic activity are well localized, we are able to derive expressions for the cortical magnification factor, the point-spread resolution, and the bandwidth resolution of the map. As a highlight, we show analytically that the receptive field size of a cortical neuron in the map is inversely proportional to the cortical magnification factor at that map location, the experimentally well-established rule of inverse magnification in retinotopic and somatotopic maps.

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7. A Tree-Structured Algorithm for Reducing Computation in Networks with Separable Basis Functions

Terence D. Sanger

https://doi.org/10.1162/neco.1991.3.1.67 | pp. 67-78

I describe a new algorithm for approximating continuous functions in high-dimensional input spaces. The algorithm builds a tree-structured network of variable size, which is determined both by the distribution of the input data and by the function to be approximated. Unlike other tree-structured algorithms, learning occurs through completely local mechanisms and the weights and structure are modified incrementally as data arrives. Efficient computation in the tree structure takes advantage of the potential for low-order dependencies between the output and the individual dimensions of the input. This algorithm is related to the ideas behind k-d trees (Bentley 1975), CART (Breiman et al. 1984), and MARS (Friedman 1988). I present an example that predicts future values of the Mackey-Glass differential delay equation.

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8. Adaptive Mixtures of Local Experts

Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, Geoffrey E. Hinton

https://doi.org/10.1162/neco.1991.3.1.79 | pp. 79-87

We present a new supervised learning procedure for systems composed of many separate networks, each of which learns to handle a subset of the complete set of training cases. The new procedure can be viewed either as a modular version of a multilayer supervised network, or as an associative version of competitive learning. It therefore provides a new link between these two apparently different approaches. We demonstrate that the learning procedure divides up a vowel discrimination task into appropriate subtasks, each of which can be solved by a very simple expert network.

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9. Efficient Training of Artificial Neural Networks for Autonomous Navigation

Dean Pomerleau

https://doi.org/10.1162/neco.1991.3.1.88 | pp. 88-97

The ALVINN (Autonomous Land Vehicle In a Neural Network) project addresses the problem of training artificial neural networks in real time to perform difficult perception tasks. ALVINN is a backpropagation network designed to drive the CMU Navlab, a modified Chevy van. This paper describes the training techniques that allow ALVINN to learn in under 5 minutes to autonomously control the Navlab by watching the reactions of a human driver. Using these techniques, ALVINN has been trained to drive in a variety of circumstances including single-lane paved and unpaved roads, and multilane lined and unlined roads, at speeds of up to 20 miles per hour.

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10. Sequence Manipulation Using Parallel Mapping Networks

David S. Touretzky, Deirdre Wheeler

https://doi.org/10.1162/neco.1991.3.1.98 | pp. 98-109

We describe a parallel mapping matrix that performs several types of sequence manipulations that are the building blocks of well-known phonological processes. Our results indicate that human phonological behavior can by modeled by a highly constrained connectionist architecture, one that uses purely feedforward circuitry and imposes tight limits on depth of derivations.

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11. Parsing Complex Sentences with Structured Connectionist Networks

Ajay N. Jain

https://doi.org/10.1162/neco.1991.3.1.110 | pp. 110-120

A modular, recurrent connectionist network is taught to incrementally parse complex sentences. From input presented one word at a time, the network learns to do semantic role assignment, noun phrase attachment, and clause structure recognition, for sentences with both active and passive constructions and center-embedded clauses. The network makes syntactic and semantic predictions at every step. Previous predictions are revised as expectations are confirmed or violated with the arrival of new information. The network induces its own "grammar rules" for dynamically transforming an input sequence of words into a syntactic/semantic interpretation. The network generalizes well and is tolerant of ill-formed inputs.

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12. Rules and Variables in Neural Nets

Venkat Ajjanagadde, Lokendra Shastri

https://doi.org/10.1162/neco.1991.3.1.121 | pp. 121-134

A fundamental problem that must be addressed by connectionism is that of creating and representing dynamic structures (Feldman 1982; von der Malsburg 1985). In the context of reasoning with systematic and abstract knowledge, this problem takes the form of the variable binding problem. We describe a biologically plausible solution to this problem and outline how a knowledge representation and reasoning system can use this solution to perform a class of predictive inferences with extreme efficiency. The proposed system solves the variable binding problem by propagating rhythmic patterns of activity wherein dynamic bindings are represented as the synchronous firing of appropriate nodes.

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13. TAG: A Neural Network Model for Large-Scale Optical Implementation

Hyuek-Jae Lee, Soo-Young Lee, Sang-Yung Shin, Bo-Yun Koh

https://doi.org/10.1162/neco.1991.3.1.135 | pp. 135-143

TAG (Training by Adaptive Gain) is a new adaptive learning algorithm developed for optical implementation of large-scale artificial neural networks. For fully interconnected single-layer neural networks with N input and M output neurons TAG contains two different types of interconnections, i.e., M N global fixed interconnections and N + M adaptive gain controls. For two-dimensional input patterns the former may be achieved by multifacet holograms, and the latter by spatial light modulators (SLMs). For the same number of input and output neurons TAG requires much less adaptive elements, and provides a possibility for large-scale optical implementation at some sacrifice in performance as compared to the perceptron. The training algorithm is based on gradient descent and error backpropagation, and is easily extensible to multilayer architecture. Computer simulation demonstrates reasonable performance of TAG compared to perceptron performance. An electrooptical implementation of TAG is also proposed.

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14. Pacemaker-Induced Coherence in Cortical Networks

Alain Destexhe, A. Babloyantz

https://doi.org/10.1162/neco.1991.3.2.145 | pp. 145-154

A simple mathematical model of cortical tissue is introduced and the system's dynamics is monitored when a small subset of neurons is submitted to oscillatory inputs of various frequency and waveform. In the absence of input, the system shows desynchronized or "turbulent" behavior. The oscillatory input synchronizes the neuronal activity, which is strongest for inputs of low frequency. The increase of spatial coherence is estimated from the spatial autocorrelation function whereas the increase in temporal coherence is evaluated from correlation dimensions. The model accounts qualitatively for some of the features of the thalamocortical system.

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15. Stimulus-Dependent Assembly Formation of Oscillatory Responses: I. Synchronization

Peter König, Thomas B. Schillen

https://doi.org/10.1162/neco.1991.3.2.155 | pp. 155-166

Current concepts in neurobiology of vision assume that local object features are represented by distributed neuronal populations in the brain. Such representations can lead to ambiguities if several distinct objects are simultaneously present in the visual field. Temporal characteristics of the neuronal activity have been proposed as a possible solution to this problem and have been found in various cortical areas. In this paper we introduce a delayed nonlinear oscillator to investigate temporal coding in neuronal networks. We show synchronization within two-dimensional layers consisting of oscillatory elements coupled by excitatory delay connections. The observed correlation length is large compared to coupling length. Following the experimental situation, we then demonstrate the response of such layers to two short stimulus bars of varying gap distance. Coherency of stimuli is reflected by the temporal correlation of the responses, which closely resembles the experimental observations.

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16. Stimulus-Dependent Assembly Formation of Oscillatory Responses: II. Desynchronization

Thomas B. Schillen, Peter König

https://doi.org/10.1162/neco.1991.3.2.167 | pp. 167-178

Recent theoretical and experimental work suggests a temporal structure of neuronal spike activity as a potential mechanism for solving the binding problem in the brain. In particular, recordings from cat visual cortex demonstrate the possibility that stimulus coherency is coded by synchronization of oscillatory neuronal responses. Coding by synchronized oscillatory activity has to avoid bulk synchronization within entire cortical areas. Recent experimental evidence indicates that incoherent stimuli can activate coherently oscillating assemblies of cells that are not synchronized among one another. In this paper we show that appropriately designed excitatory delay connections can support the desynchronization of two-dimensional layers of delayed nonlinear oscillators. Closely following experimental observations, we then present two examples of stimulus-dependent assembly formation in oscillatory layers that employ both synchronizing and desynchronizing delay connections: First, we demonstrate the segregation of oscillatory responses to two overlapping but incoherently moving stimuli. Second, we show that the coherence of movement and location of two stimulus bar segments can be coded by the correlation of oscillatory activity.

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17. Recurrent Network Model of the Neural Mechanism of Short-Term Active Memory

David Zipser

https://doi.org/10.1162/neco.1991.3.2.179 | pp. 179-193

Two decades of single unit recording in monkeys performing short-term memory tasks has established that information can be stored as sustained neural activity. The mechanism of this information storage is unknown. The learning-based model described here demonstrates that a mechanism using only the dynamic activity in recurrent networks is sufficient to account for the observed phenomena. The temporal activity patterns of neurons in the model match those of real memory-associated neurons, while the model's gating properties and attractor dynamics provide explanations for puzzling aspects of the experimental data.

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18. Learning Invariance from Transformation Sequences

Péter Földiák

https://doi.org/10.1162/neco.1991.3.2.194 | pp. 194-200

The visual system can reliably identify objects even when the retinal image is transformed considerably by commonly occurring changes in the environment. A local learning rule is proposed, which allows a network to learn to generalize across such transformations. During the learning phase, the network is exposed to temporal sequences of patterns undergoing the transformation. An application of the algorithm is presented in which the network learns invariance to shift in retinal position. Such a principle may be involved in the development of the characteristic shift invariance property of complex cells in the primary visual cortex, and also in the development of more complicated invariance properties of neurons in higher visual areas.

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19. A Biologically Supported Error-Correcting Learning Rule

Peter Hancock, Leslie S. Smith, William A. Phillips

https://doi.org/10.1162/neco.1991.3.2.201 | pp. 201-212

. 1990) can support an error-correcting learning rule. The rule increases weights when both pre- and postsynaptic units are highly active, and decreases them when pre-synaptic activity is high and postsynaptic activation is less than the threshold for weight increment but greater than a lower threshold. We show that this rule corrects false positive outputs in feedforward associative memory, that in an appropriate opponent-unit architecture it corrects misses, and that it performs better than the optimal Hebbian learning rule reported by Willshaw and Dayan (1990).

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20. A Resource-Allocating Network for Function Interpolation

John Platt

https://doi.org/10.1162/neco.1991.3.2.213 | pp. 213-225

We have created a network that allocates a new computational unit whenever an unusual pattern is presented to the network. This network forms compact representations, yet learns easily and rapidly. The network can be used at any time in the learning process and the learning patterns do not have to be

repeated. The units in this network respond to only a local region of the space of input values. The network learns by allocating new units and adjusting the parameters of existing units. If the network performs poorly on a presented pattern, then a new unit is allocated that corrects the response to the presented pattern. If the network performs well on a presented pattern, then the network parameters are updated using standard LMS gradient descent. We have obtained good results with our resource-allocating network (RAN). For predicting the Mackey-Glass chaotic time series, RAN learns much faster than do those using backpropagation networks and uses a comparable number of synapses.

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21. On the Convergence of the LMS Algorithm with Adaptive Learning Rate for Linear Feedforward Networks

Zhi■Quan Luo

https://doi.org/10.1162/neco.1991.3.2.226 | pp. 226-245

We consider the problem of training a linear feedforward neural network by using a gradient descent-like LMS learning algorithm. The objective is to find a weight matrix for the network, by repeatedly presenting to it a finite set of examples, so that the sum of the squares of the errors is minimized. Kohonen showed that with a small but fixed learning rate (or stepsize) some subsequences of the weight matrices generated by the algorithm will converge to certain matrices close to the optimal weight matrix. In this paper, we show that, by dynamically decreasing the learning rate during each training cycle, the sequence of matrices generated by the algorithm will converge to the optimal weight matrix. We also show that for any given ■ > 0 the LMS algorithm, with decreasing learning rates, will generate an ■-optimal weight matrix (i.e., a matrix of distance at most ■ away from the optimal matrix) after O(1/■) training cycles. This is in contrast to Ω(1/■log 1/■) training cycles needed to generate an ■-optimal weight matrix when the learning rate is kept fixed. We also give a general condition for the learning rates under which the LMS learning algorithm is guaranteed to converge to the optimal weight matrix.

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22. Universal Approximation Using Radial-Basis-Function Networks

Jihun Park, Irwin W. Sandberg

https://doi.org/10.1162/neco.1991.3.2.246 | pp. 246-257

There have been several recent studies concerning feedforward networks and the problem of approximating arbitrary functionals of a finite number of real variables. Some of these studies deal with cases in which the hidden-layer nonlinearity is not a sigmoid. This was motivated by successful applications of feedforward networks with nonsigmoidal hidden-layer units. This paper reports on a related study of radial-basis-function (RBF) networks, and it is proved that RBF networks having one hidden layer are capable of universal approximation. Here the emphasis is on the case of typical RBF networks, and the results show that a certain class of RBF networks with the same smoothing factor in each kernel node is broad enough for universal approximation.

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23. Recognizing Hand-Printed Letters and Digits Using Backpropagation Learning

Gale L. Martin, James A. Pittman
https://doi.org/10.1162/neco.1991.3.2.258 | pp. 258-267

We report on results of training backpropagation nets with samples of hand-printed digits scanned off of bank checks and hand-printed letters interactively entered into a computer through a stylus digitizer. Generalization results are reported as a function of training set size and network capacity. Given a large training set, and a net with sufficient capacity to achieve high performance on the training set, nets typically achieved error rates of 4-5% at a 0% reject rate and 1-2% at a 10% reject rate. The topology and capacity of the system, as measured by the number of connections in the net, have surprisingly little effect on generalization. For those developing hand-printed character recognition systems, these results suggest that a large and representative training sample may be the single, most important factor in achieving high recognition accuracy. Benefits of reducing the number of net connections, other than improving generalization, are discussed.

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24. Constrained Nets for Graph Matching and Other Quadratic Assignment Problems

Petar Simi■
https://doi.org/10.1162/neco.1991.3.2.268 | pp. 268-281

Some time ago Durbin and Willshaw proposed an interesting parallel algorithm (the "elastic net") for approximately solving some geometric optimization problems, such as the Traveling Salesman Problem. Recently it has been shown that their algorithm is related to neural networks of Hopfield and Tank, and that they both can be understood as the semiclassical approximation to statistical mechanics of related physical models. The main point of the elastic net algorithm is seen to be in the way one deals with the constraints when evaluating the effective cost function (free energy in the thermodynamic analogy), and not in its geometric foundation emphasized originally by Durbin and Willshaw. As a consequence, the elastic net algorithm is a special case of the more general physically based computations and can be generalized to a large class of nongeometric problems. In this paper we further elaborate on this observation, and generalize the elastic net to the quadratic assignment problem. We work out in detail its special case, the graph matching problem, because it is an important problem with many applications in computational vision and neural modeling. Simulation results on random graphs, and on structured (hand-designed) graphs of moderate size (20-100 nodes) are discussed.

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25. Symmetric Neural Networks and Propositional Logic Satisfiability

Gadi Pinkas
https://doi.org/10.1162/neco.1991.3.2.282 | pp. 282-291

Connectionist networks with symmetric weights (like Hopfield networks and Boltzmann Machines) use gradient descent to find a minimum for quadratic energy functions. We show equivalence between the problem of satisfiability in propositional calculus and the problem of minimizing those energy functions. The equivalence is in the sense that for any satisfiable well-formed formula (WFF) we can find a quadratic function that describes it, such that the set of solutions that minimizes the function is equal to the set of truth assignments that satisfy the WFF. We also show that in the same sense every quadratic energy function

describes some satisfiable WFF. Algorithms are given to transform any propositional WFF into an energy function that describes it and vice versa. High-order models that use sigma-pi units are shown to be equivalent to the standard quadratic models with additional hidden units. An algorithm to convert high-order networks to low-order ones is used to implement a satisfiability problem-solver on a connectionist network. The results give better understanding of the role of hidden units and of the limitations and capabilities of symmetric connectionist models. The techniques developed for the satisfiability problem may be applied to a wide range of other problems, such as associative memories, finding maximal consistent subsets, automatic deduction, and even nonmonotonic reasoning.

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26. A Practical Approach for Representing Context and for Performing Word Sense Disambiguation Using Neural Networks

Stephen I. Gallant

https://doi.org/10.1162/neco.1991.3.3.293 | pp. 293-309

Representing and manipulating context information is one of the hardest problems in natural language processing. This paper proposes a method for representing some context information so that the correct meaning for a word in a sentence can be selected. The approach is primarily based on work by Waltz and Pollack (1985, 1984), who emphasized neutrally plausible systems. By contrast this paper focuses on computationally feasible methods applicable to full-scale natural language processing systems. There are two key elements: a collection of context vectors defined for every word used by a natural language processing system, and a context algorithm that computes a dynamic context vector at any position in a body of text. Once the dynamic context vector has been computed it is easy to choose among competing meanings for a word. This choice of definitions is essentially a neural network computation, and neural network learning algorithms should be able to improve such choices. Although context vectors do not represent all context information, their use should improve those full-scale systems that have avoided context as being too difficult to deal with. Good candidates for full-scale context vector implementations are machine translation systems and Japanese word processors. A main goal of this paper is to encourage such large-scale implementations and tests of context vector approaches. A variety of interesting directions for research in natural language processing and machine learning will be possible once a full set of context vectors has been created. In particular the development of more powerful context algorithms will be an important topic for future research.

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27. A Modified Quickprop Algorithm

Alistair Veitch, Geoffrey Holmes

https://doi.org/10.1162/neco.1991.3.3.310 | pp. 310-311

September 01 1991 A Modified Quickprop Algorithm Alistair Craig Veitch, Alistair Craig Veitch Department of Computer Science, University of Waikato, New Zealand Search for other works by this author on: This Site Google Scholar Geoffrey Holmes Geoffrey Holmes Department of Computer Science, University of Waikato, New Zealand Search for other works by this author on: This Site Google Scholar Author and Article Information Alistair Craig Veitch Department of Computer Science, University of Waikato, New Zealand Geoffrey Holmes Department of Computer Science, University of Waikato, New Zealand Received: February 11 1991 Accepted: March 04 1991 Online Issn: 1530-888X Print Issn: 0899-7667 © 1991

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28. Removing Time Variation with the Anti-Hebbian Differential Synapse

Graeme Mitchison

https://doi.org/10.1162/neco.1991.3.3.312 | pp. 312-320

I describe a local synaptic learning rule that can be used to remove the effects of certain types of systematic temporal variation in the inputs to a unit. According to this rule, changes in synaptic weight result from a conjunction of short-term temporal changes in the inputs and the output. Formally, This is like the differential rule proposed by Klopf (1986) and Kosko (1986), except for a change of sign, which gives it an anti-Hebbian character. By itself this rule is insufficient. A weight conservation condition is needed to prevent the weights from collapsing to zero, and some further constraint-implemented here by a biasing term-to select particular sets of weights from the subspace of those which give minimal variation. As an example, I show that this rule will generate center-surround receptive fields that remove temporally varying linear gradients from the inputs.

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29. Simulations of a Reconstructed Cerebellar Purkinje Cell Based on Simplified Channel Kinetics

Paul C. Bush, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1991.3.3.321 | pp. 321-332

When cerebellar Purkinje cells are depolarized with a constant current pulse injected at the soma, complex spike discharge patterns are observed (Llinas and Sugimori 1980b). A computer model has been constructed to analyze how the Purkinje cell ionic conductance identified to date interact to produce the observed firing behavior. The kinetics of voltage-dependent conductance used in the model were significantly simpler than Hodgkin-Huxley kinetics, which have many parameters that must be experimentally determined. Our simplified scheme was able to reproduce the complex nonlinear responses found in real Purkinje cells. A similar approach could be used to study the wide variety of neurons found in different brain regions.

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30. On the Mechanisms Underlying Directional Selectivity

Haluk Ö■men

https://doi.org/10.1162/neco.1991.3.3.333 | pp. 333-349

Recent efforts in the understanding of motion detection and directional selectivity include electrophysiological studies using single photoreceptor stimulations and a combination of electrophysiology and neuropharmacology. Results of the former have been interpreted in favor of facilitator motion detection models while results of the latter have been interpreted in favor of inhibitory models. In this paper, this conflicting data interpretation problem is addressed by mathematically modeling some effects of neuropharmacological substances and by applying this formalism to a neural network model of directionally selective motion perception. The study offers a possible resolution to the paradox.

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31. 2-Degree-of-freedom Robot Path Planning using Cooperative Neural Fields

Michael Lemmon

https://doi.org/10.1162/neco.1991.3.3.350 | pp. 350-362

This paper proposes a neural network solution to path planning by two degree-of-freedom (DOF) robots. The proposed network is a two-dimensional sheet of neurons forming a distributed representation of the robot's workspace. Lateral interconnections between neurons are "cooperative," so that the field exhibits oscillatory behavior. This paper shows how that oscillatory behavior can be used to solve the path-planning problem. The results reported show that the proposed neural network finds the variational solution of Bellman's dynamic programming equation.

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32. Parameter Sensitivity of the Elastic Net Approach to the Traveling Salesman Problem

Martin W. Simmen

https://doi.org/10.1162/neco.1991.3.3.363 | pp. 363-374

Durbin and Willshaw's elastic net algorithm can find good solutions to the TSP. The purpose of this paper is to point out that for certain ranges of parameter values, the algorithm converges into local minima that do not correspond to valid tours. The key parameter is the ratio governing the relative strengths of the two competing terms in the elastic net energy function. Based on recent work by Durbin, Szeliski and Yuille, the parameter regime in which the net may visit some cities twice is examined. Further analysis predicts the regime in which the net may fail to visit some cities at all. Understanding these limitations allows one to select the parameter value most likely to avoid either type of problem. Simulation data support the theoretical work.

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33. FIR and IIR Synapses, a New Neural Network Architecture for Time Series Modeling

Andrew D. Back, Ah Chung Tsoi

https://doi.org/10.1162/neco.1991.3.3.375 | pp. 375-385

A new neural network architecture involving either local feedforward global feedforward, and/or local recurrent global feedforward structure is proposed. A learning rule minimizing a mean square error criterion is derived. The performance of this algorithm (local recurrent global feedforward architecture) is compared with a local-feedforward global-feedforward architecture. It is shown that the local-recurrent global-feedforward model performs better than the local-feedforward global-feedforward model.

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34. The Transition to Perfect Generalization in Perceptrons

Eric B. Baum, Yuh■Dauh Lyuu

https://doi.org/10.1162/neco.1991.3.3.386 | pp. 386-401

Several recent papers (Gardner and Derrida 1989; Györgyi 1990; Sompolinsky et al. 1990) have found, using methods of statistical physics, that a transition to perfect generalization occurs in training a simple perceptron whose weights can only take values ±1. We give a rigorous proof of such a phenomena. That is, we show, for $\alpha = 2.0821$, that if at least αn examples are drawn from the uniform distribution on $\{+1, -1\}^n$ and classified according to a target perceptron $w_t \in \{+1, -1\}^n$ as positive or negative according to whether $w_t \cdot x$ is nonnegative or negative, then the probability is $2^{-(\sqrt{n})}$ that there is any other such perceptron consistent with the examples. Numerical results indicate further that perfect generalization holds for α as low as 1.5.

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35. Learning by Asymmetric Parallel Boltzmann Machines

Bruno Apolloni, Diego de Falco

https://doi.org/10.1162/neco.1991.3.3.402 | pp. 402-408

We consider the Little, Shaw, Vasudevan model as a parallel asymmetric Boltzmann machine, in the sense that we extend to this model the entropic learning rule first studied by Ackley, Hinton, and Sejnowski in the case of a sequentially activated network with symmetric synaptic matrix. The resulting Hebbian learning rule for the parallel asymmetric model draws the signal for the updating of synaptic weights from time averages of the discrepancy between expected and actual transitions along the past history of the network. As we work without the hypothesis of symmetry of the weights, we can include in our analysis also feedforward networks, for which the entropic learning rule turns out to be complementary to the error backpropagation rule, in that it "rewards the correct behavior" instead of "penalizing the wrong answers."

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36. Generalization Effects of k-Neighbor Interpolation Training

Takeshi Kawabata

https://doi.org/10.1162/neco.1991.3.3.409 | pp. 409-417

This paper describes a new training method for a continuous mapping and/or pattern classification neural network that performs local sample-density smoothing. A conventional training method uses point-to-point mapping from an input space to an output space. Even though the mapping may be precise at two given training sample points, there are no guarantees of mapping accuracy at points on a line segment connecting the sample points. This paper first discusses a theory for formulating line-to-line mapping. The theory is called interpolation training. This paper then expands the theory to k-nearest neighbor interpolation. The

k-neighbor interpolation training (KNIT) method connects an input sample training point to its k-neighbor points via k line segments. Then, the method maps these k line segments in the input space for each training sample to linear line segments in the output space that interpolate between training output values. Thus, a web structure made by connecting input samples is mapped into the same structure in an output space. The KNIT method reduces the over learning problem caused by point-to-point training by smoothing input/output functions. Simulation tasks show that KNIT improves vowel recognition on a small speech database.

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37. Including Hints in Training Neural Nets

Khalid Al-Mashouq, I.S. Reed

https://doi.org/10.1162/neco.1991.3.3.418 | pp. 418-427

The aim of a neural net is to partition the data space into near optimal decision regions. Learning such a partitioning solely from examples has proven to be a very hard problem (Blum and Rivest 1988; Judd 1988). To remedy this, we use the idea of supplying hints to the network-as discussed by Abu-Mostafa (1990). Hints reduce the solution space, and as a consequence speed up the learning process. The minimum Hamming distance between the patterns serves as the hint. Next, it is shown how to learn such a hint and how to incorporate it into the learning algorithm. Modifications in the net structure and its operation are suggested, which allow for a better generalization. The sensitivity to errors in such a hint is studied through some simulations.

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38. On the Characteristics of the Autoassociative Memory with Nonzero-Diagonal Terms in the Memory Matrix

Jung-Hua Wang, Thomas F. Krile, John F. Walkup, Tai■Lang Jong

https://doi.org/10.1162/neco.1991.3.3.428 | pp. 428-439

A statistical method is applied to explore the unique characteristics of a certain class of neural network autoassociative memory with N neurons and first-order synaptic interconnections. The memory matrix is constructed to store M = αN vectors based on the outer-product learning algorithm. We theoretically prove that, by setting all the diagonal terms of the memory matrix to be M and letting the input error ratio ρ = 0, the probability of successful recall P r steadily decreases as α increases, but as α increases past 1.0, P r begins to increase slowly. When 0 < ρ ≤ 0.5, the network exhibits strong error-correction capability if α ≤ 0.15 and this capability is shown to rapidly decrease as α increases. The network essentially loses all its error-correction capability at α = 2, regardless of the value of ρ. When 0 < ρ ≤ 0.5, and under the constraint of P r > 0.99, the tradeoff between the number of stable states and their attraction force is analyzed and the memory capacity is shown to be 0.15N at best.

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39. Handwritten Digit Recognition Using K Nearest-Neighbor, Radial-Basis Function, and Backpropagation Neural Networks

Yuchun Lee

https://doi.org/10.1162/neco.1991.3.3.440 | pp. 440-449

Results of recent research suggest that carefully designed multilayer neural networks with local “receptive fields” and shared weights may be unique in providing low error rates on handwritten digit recognition tasks. This study, however, demonstrates that these networks, radial basis function (RBF) networks, and k nearest-neighbor (kNN) classifiers, all provide similar low error rates on a large handwritten digit database. The backpropagation network is overall superior in memory usage and classification time but can provide “false positive” classifications when the input is not a digit. The backpropagation network also has the longest training time. The RBF classifier requires more memory and more classification time, but less training time. When high accuracy is warranted, the RBF classifier can generate a more effective confidence judgment for rejecting ambiguous inputs. The simple kNN classifier can also perform handwritten digit recognition, but requires a prohibitively large amount of memory and is much slower at classification. Nevertheless, the simplicity of the algorithm and fast training characteristics makes the kNN classifier an attractive candidate in hardware-assisted classification tasks. These results on a large, high input dimensional problem demonstrate that practical constraints including training time, memory usage, and classification time often constrain classifier selection more strongly than small differences in overall error rate.

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40. A Matrix Method for Optimizing a Neural Network

S.A. Barton

https://doi.org/10.1162/neco.1991.3.3.450 | pp. 450-459

A matrix method is described that optimizes the set of weights and biases for the output side of a network with a single hidden layer of neurons, given any set of weights and biases for the input side of the hidden layer. All the input patterns are included in a single optimization cycle. A simple iterative minimization procedure is used to optimize the weights and biases on the input side of the hidden layer. Many test problems have been solved, confirming the validity of the method. The results suggest that for a network with a single layer of hidden sigmoidal nodes, the accuracy of a functional representation is reduced as the nonlinearity of the function increases.

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41. Neural Network Classifiers Estimate Bayesian a posteriori Probabilities

Michael D. Richard, Richard P. Lippmann

https://doi.org/10.1162/neco.1991.3.4.461 | pp. 461-483

Many neural network classifiers provide outputs which estimate Bayesian a posteriori probabilities. When the estimation is accurate, network outputs can be treated as probabilities and sum to one. Simple proofs show that Bayesian probabilities are estimated when desired network outputs are 1 of M (one output unity, all others zero) and a squared-error or cross-entropy cost function is used. Results of Monte Carlo simulations performed using multilayer perceptron (MLP) networks trained with backpropagation, radial basis function (RBF) networks, and high-order polynomial networks graphically demonstrate that network outputs provide good estimates of Bayesian probabilities. Estimation accuracy depends on network complexity, the amount of training data, and the degree to which training data reflect true likelihood distributions and a priori class probabilities. Interpretation of network outputs as Bayesian probabilities allows outputs from multiple networks to be combined for higher level decision making, simplifies creation of

rejection thresholds, makes it possible to compensate for differences between pattern class probabilities in training and test data, allows outputs to be used to minimize alternative risk functions, and suggests alternative measures of network performance.

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42. Lowering Variance of Decisions by Using Artificial Neural Network Portfolios

Ganesh Mani

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December 01 1991 Lowering Variance of Decisions by Using Artificial Neural Network Portfolios In Special Collection: CogNet Ganesh Mani Ganesh Mani Computer Sciences Department, University of Wisconsin—Madison, 1210 W. Dayton Street, Madison, WI 53706 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Ganesh Mani Computer Sciences Department, University of Wisconsin—Madison, 1210 W. Dayton Street, Madison, WI 53706 USA Received: February 06 1991 Accepted: May 03 1991 Online Issn: 1530-888X Print Issn: 0899-7667 © 1991 Massachusetts Institute of Technology1991 Neural Computation (1991) 3 (4): 484–486. https://doi.org/10.1162/neco.1991.3.4.484 Article history Received: February 06 1991 Accepted: May 03 1991 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Ganesh Mani; Lowering Variance of Decisions by Using Artificial Neural Network Portfolios. Neural Comput 1991; 3 (4): 484–486. doi: https://doi.org/10.1162/neco.1991.3.4.484 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1991 Massachusetts Institute of Technology1991 Article PDF first page preview Close Modal You do not currently have access to this content.

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43. Oscillating Networks: Control of Burst Duration by Electrically Coupled Neurons

L. F. Abbott, Eve Marder, Scott L. Hooper

https://doi.org/10.1162/neco.1991.3.4.487 | pp. 487-497

The pyloric network of the stomatogastric ganglion in crustacea is a central pattern generator that can produce the same basic rhythm over a wide frequency range. Three electrically coupled neurons, the anterior burster (AB) neuron and two pyloric dilator (PD) neurons, act as a pacemaker unit for the pyloric network. The functional characteristics of the pacemaker network are the result of electrical coupling between neurons with quite different intrinsic properties, each contributing a basic feature to the complete circuit. The AB neuron, a conditional oscillator, plays a dominant role in rhythm generation. In the work described here, we manipulate the frequency of the AB neuron both isolated and electrically coupled to the PD neurons. Physiological and modeling studies indicate that the PD neurons play an important role in regulating the duration of the bursts produced by the pacemaker unit.

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44. A Computer Simulation of Oscillatory Behavior in Primary Visual Cortex

Matthew A. Wilson, James M. Bower

https://doi.org/10.1162/neco.1991.3.4.498 | pp. 498-509

Periodic variations in correlated cellular activity have been observed in many regions of the cerebral cortex. The recent discovery of stimulus-dependent, spatially-coherent oscillations in primary visual cortex of the cat has led to suggestions of neural information encoding schemes based on phase and/or frequency variation. To explore the mechanisms underlying this behavior and their possible functional consequences, we have developed a realistic neural model, based on structural features of visual cortex, which replicates observed oscillatory phenomena. In the model, this oscillatory behavior emerges directly from the structure of the cortical network and the properties of its intrinsic neurons; however, phase coherence is shown to be an average phenomenon seen only when measurements are made over multiple trials. Because average coherence does not ensure synchrony of firing over the course of single stimuli, oscillatory phase may not be a robust strategy for directly encoding stimulus-specific information. Instead, the phase and frequency of cortical oscillations may reflect the coordination of general computational processes within and between cortical areas. Under this interpretation, coherence emerges as a result of horizontal interactions that could be involved in the formation of receptive field properties.

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45. Segmentation, Binding, and Illusory Conjunctions

D. Horn, Dov Sagi, Marius Usher

https://doi.org/10.1162/neco.1991.3.4.510 | pp. 510-525

We investigate binding within the framework of a model of excitatory and inhibitory cell assemblies that form an oscillating neural network. Our model is composed of two such networks that are connected through their inhibitory neurons. The excitatory cell assemblies represent memory patterns. The latter have different meanings in the two networks, representing two different attributes of an object, such as shape and color. The networks segment an input that contains mixtures of such pairs into staggered oscillations of the relevant activities. Moreover, the phases of the oscillating activities representing the two attributes in each pair lock with each other to demonstrate binding. The system works very well for two inputs, but displays faulty correlations when the number of objects is larger than two. In other words, the network conjoins attributes of different objects, thus showing the phenomenon of "illusory conjunctions," as in human vision.

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46. Contrastive Learning and Neural Oscillations

Pierre Baldi, Fernando J. Pineda

https://doi.org/10.1162/neco.1991.3.4.526 | pp. 526-545

The concept of Contrastive Learning (CL) is developed as a family of possible learning algorithms for neural networks. CL is an extension of Deterministic Boltzmann Machines to more general dynamical systems. During learning, the network oscillates between two phases. One phase has a teacher signal and one phase has no teacher signal. The weights are updated using a learning rule that corresponds to gradient descent on a contrast function that measures the discrepancy between the free network and the network with a teacher signal. The CL approach provides a general unified framework for developing new learning algorithms. It also shows that many different types of clamping and teacher signals are possible. Several

examples are given and an analysis of the landscape of the contrast function is proposed with some relevant predictions for the CL curves. An approach that may be suitable for collective analog implementations is described. Simulation results and possible extensions are briefly discussed together with a new conjecture regarding the function of certain oscillations in the brain. In the appendix, we also examine two extensions of contrastive learning to time-dependent trajectories.

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47. Weight Perturbation: An Optimal Architecture and Learning Technique for Analog VLSI Feedforward and Recurrent Multilayer Networks

Marwan A. Jabri, B. Flower

https://doi.org/10.1162/neco.1991.3.4.546 | pp. 546-565

Previous work on analog VLSI implementation of multilayer perceptrons with on-chip learning has mainly targeted the implementation of algorithms like backpropagation. Although backpropagation is efficient, its implementation in analog VLSI requires excessive computational hardware. In this paper we show that, for analog parallel implementations, the use of gradient descent with direct approximation of the gradient using "weight perturbation" instead of backpropagation significantly reduces hardware complexity. Gradient descent by weight perturbation eliminates the need for derivative and bidirectional circuits for on-chip learning, and access to the output states of neurons in hidden layers for off-chip learning. We also show that weight perturbation can be used to implement recurrent networks. A discrete level analog implementation showing the training of an XOR network as an example is described.

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48. Predicting the Future: Advantages of Semilocal Units

Eric Hartman, James D. Keeler

https://doi.org/10.1162/neco.1991.3.4.566 | pp. 566-578

In investigating gaussian radial basis function (RBF) networks for their ability to model nonlinear time series, we have found that while RBF networks are much faster than standard sigmoid unit backpropagation for low-dimensional problems, their advantages diminish in high-dimensional input spaces. This is particularly troublesome if the input space contains irrelevant variables. We suggest that this limitation is due to the localized nature of RBFs. To gain the advantages of the highly nonlocal sigmoids and the speed advantages of RBFs, we propose a particular class of semilocal activation functions that is a natural interpolation between these two families. We present evidence that networks using these gaussian bar units avoid the slow learning problem of sigmoid unit networks, and, very importantly, are more accurate than RBF networks in the presence of irrelevant inputs. On the Mackey-Glass and Coupled Lattice Map problems, the speedup over sigmoid networks is so dramatic that the difference in training time between RBF and gaussian bar networks is minor. Gaussian bar architectures that superpose composed gaussians (gaussians-of-gaussians) to approximate the unknown function have the best performance. We postulate that an interesing behavior displayed by gaussian bar functions under gradient descent dynamics, which we call automatic connection pruning, is an important factor in the success of this representation.

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49. Improving the Generalization Properties of Radial Basis Function Neural Networks

Chris Bishop

https://doi.org/10.1162/neco.1991.3.4.579 | pp. 579-588

An important feature of radial basis function neural networks is the existence of a fast, linear learning algorithm in a network capable of representing complex nonlinear mappings. Satisfactory generalization in these networks requires that the network mapping be sufficiently smooth. We show that a modification to the error functional allows smoothing to be introduced explicitly without significantly affecting the speed of training. A simple example is used to demonstrate the resulting improvement in the generalization properties of the network.

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50. Temporal Evolution of Generalization during Learning in Linear Networks

Pierre Baldi, Yves Chauvin

https://doi.org/10.1162/neco.1991.3.4.589 | pp. 589-603

We study generalization in a simple framework of feedforward linear networks with n inputs and n outputs, trained from examples by gradient descent on the usual quadratic error function. We derive analytical results on the behavior of the validation function corresponding to the LMS error function calculated on a set of validation patterns. We show that the behavior of the validation function depends critically on the initial conditions and on the characteristics of the noise. Under certain simple assumptions, if the initial weights are sufficiently small, the validation function has a unique minimum corresponding to an optimal stopping time for training for which simple bounds can be calculated. There exists also situations where the validation function can have more complicated and somewhat unexpected behavior such as multiple local minima (at most n) of variable depth and long but finite plateau effects. Additional results and possible extensions are briefly discussed.

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51. Learning the Unlearnable

Dan Nabutovsky, Eytan Domany

https://doi.org/10.1162/neco.1991.3.4.604 | pp. 604-616

We present a local perceptron-learning rule that either converges to a solution, or establishes linear nonseparability. We prove that when no solution exists, the algorithm detects this in a finite time (number of learning steps). This time is polynomial in typical cases and exponential in the worst case, when the set of patterns is nonstrictly linearly separable. The algorithm is local and has no arbitrary parameters.

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52. Kolmogorov's Theorem Is Relevant

V■ra K■rková

https://doi.org/10.1162/neco.1991.3.4.617 | pp. 617-622

We show that Kolmogorov's theorem on representations of continuous functions of n-variables by sums and superpositions of continuous functions of one variable is relevant in the context of neural networks. We give

a version of this theorem with all of the one-variable functions approximated arbitrarily well by linear combinations of compositions of affine functions with some given sigmoidal function. We derive an upper estimate of the number of hidden units.

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53. An Exponential Response Neural Net

Shlomo Geva, Joaquin Sitte

https://doi.org/10.1162/neco.1991.3.4.623 | pp. 623-632

By using artificial neurons with exponential transfer functions one can design perfect autoassociative and heteroassociative memory networks, with virtually unlimited storage capacity, for real or binary valued input and output. The autoassociative network has two layers: input and memory, with feedback between the two. The exponential response neurons are in the memory layer. By adding an encoding layer of conventional neurons the network becomes a heteroassociator and classifier. Because for real valued input vectors the dot-product with the weight vector is no longer a measure for similarity, we also consider a euclidean distance based neuron excitation and present Lyapunov functions for both cases. The network has energy minima corresponding only to stored prototype vectors. The exponential neurons make it simpler to build fast adaptive learning directly into classification networks that map real valued input to any class structure at its output.

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54. Errata

(authors unknown)

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55. Index to Volume 3

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